AI Engineer

Step 1: Learn Python & Mathematics

CareerCompass — Detailed Learning Notes

What this step is about

Build the numerical programming and mathematical foundation required for AI work.

Core Concepts

- Master Python functions, classes, modules, exceptions, virtual environments and package management.
- Learn NumPy arrays and vectorized operations.
- Study vectors, matrices, dot products, probability, statistics, derivatives and optimization intuition.
- Focus on understanding how the math maps to models instead of memorizing formulas.

Skills You Should Be Able To Demonstrate

- Use NumPy arrays and matrix operations.
- Explain mean, variance and probability.
- Explain a gradient as a direction of change.
- Implement a small numerical algorithm.

Hands-on Project / Practice

- Implement linear regression from scratch with NumPy.
- Compare it with a library implementation and explain the differences.

Recommended References

- Python Documentation — https://docs.python.org/3/
- DeepLearning.AI — https://www.deeplearning.ai/
- TensorFlow Documentation — https://www.tensorflow.org/learn